Shepherd — Presenter Brief

For the teammate writing the pitch & slides

natHacks 2026 · Standard Division · **Defense & Security**

Tailored to the handbook Evaluation rubric + submission rules

One-liner (memorize): Shepherd is a live evacuation system for Edmonton that routes people to safe shelters on real roads as a wildfire or tornado spreads with the wind — and keeps re-routing them when roads jam or become unsafe.

15-second pitch: When disaster hits Edmonton, Shepherd gives emergency managers a live command map and every resident a phone route to the nearest safe shelter. As the hazard grows with the wind and traffic piles up, we close roads early, estimate who needs to leave, and push mid-trip reroutes so people don't drive into fire or gridlock.

1 Why we built this (the problem)

Static evacuation advice ("leave west", a PDF map, a one-time SMS) fails when **fire, wind, and traffic change minute by minute**. People clog the same arteries; some shelters become unsafe; phones don't update; commanders can't see closed roads or cut-offs in one place.

Defense & Security fit (theme language from NAT): emergency response, public safety, resilient infrastructure, situational awareness. Shepherd is a **decision-support + public routing** layer for urban evacuation — not a firefighting prediction product and not certified life-safety software (say that honestly if asked).

2 Vision & what we actually shipped

2.1 V

ision One system that (1) sees the hazard grow with real wind/forecast, (2) knows Edmonton's **real OpenStreetMap roads**, (3) gives each person a **time-aware** route to an open shelter, (4) **re-routes live** when roads close, shelters go unsafe, or traffic jams, (5) shows commanders **closed roads, crowd size, live phone paths, and who's cut off next**.

2.2 T

hree surfaces	**Dashboard** (:5173)	Commanders: wildfire/tornado controls, wind, forecast rings, traffic ribbons, fire-closed roa
		phone dots + blue live routes, cut-off banner, QR to join.
	Phone (:5174)	Public/judges: route options, leave-by countdown, simulated drive (default) or Real GPS, mid-t
		reroutes, hazard reports, priority/help. Starts from their real GPS pin, then they can move it.
	API (:8000)	FastAPI, in-memory state. Contract `shared/contract.md` v2.1.

2.3 C

ore product behaviours (say these in the demo)

1. **Wind-driven hazard:** wildfire or tornado; wind speed + FROM direction; sim clock `T+N`; forecast outlines (e.g. T+60).
2. **Road safety rule:** close a road at `arrival_min − 30` (30 min before fire is predicted to arrive). Route statuses: **ok** / **tight** / **blocked**.
3. **Multi-option routing** on real OSM (OSMnx/NetworkX + SciPy Dijkstra), scored against fire timing + traffic delay.
4. **Live phone reroutes** while evacuating (`shouldReroute`); dashboard shows blue path redraw.
5. **Traffic:** BPR volume–delay, anti-gridlock assignment, projected crowd up to ∼**100k cars**, demand from pin + Edmonton zone density + day/time.
6. **Commander overlays:** closed roads, congestion ribbons, cut-off warnings, Compare (naive pile-up vs spread clearance).
7. **Demo join:** `./demo.sh` + Cloudflare tunnels + QR so judges' phones work off venue Wi-Fi.

3 Handbook scoring map (use this to structure slides)

Source: natHacks 2026 Hacker Handbook **EVALUATION** page ("VIEW THE FULL RUBRIC" link may have deeper detail — this table is the official category list).

Category	Max	How Shepherd scores it — what to say / show
Technical Execution & Problem-Solving	10	Real OSM graph; wind + `arrival_min`; timed road closures; BPR traffic + projection so large N stays real-time; shared API contract across 5 builders; FastAPI + React + MapLibre. Show closed roads + reroute loop.
Prototype Functionality & Usability	10	End-to-end live demo: dashboard + phone on QR. Leave-by countdown, route cards, sim drive default, map taps, reports. Ops-style blue/white UI (Public Sans). Reset cleans map.
Creativity & Innovation	10	Not "another map." The closed loop: **wind hazard → timed closures → mid-trip phone reroute → commander cut-offs + demand** on real Edmonton streets. Anti-gridlock vs naive pile-up Compare.
Impact & Real-World Relevance	10	Alberta wildfire/evac reality; Edmonton shelters; public safety + emergency ops. Honest: prototype, not certified. Path to AES/municipal ops + open census demand.
Feasibility & Continuation Potential	10	Runs today with `./demo.sh`. Next: census demand, WebSockets, agency auth/roles, live shelter intake, calibrated fire models, offline phone cache. Seed funding = validate with emergency managers.
Pitch & Demonstration	10	5 min present/demo + 5 min Q&A (handbook). Script below. Video OK if inside the 5 min. All teammates attend judging unless excused.
Written Summary	5	Paste the 250-word summary below into the submission form.
Division-Specific Bonus	12	Standard Division: team composition / interdisciplinary bonus. Emphasize 5 roles: API/integration, routing, sim, dashboard, phone — software + product + ops UX.

4 Submission checklist (handbook)

- Deadline: **Nov 9, 11:00 AM MDT** via Discord form (late not accepted). Handbook submissions page; confirm exact date on Discord if it differs.
- **Project summary** $\leq$ 250 words (draft below).
- **About the project** $\leq$ 500 words: inspiration, learnings, how built, challenges (draft below).
- **Code repo** built during the 64-hour weekend + README (run instructions, links, credits).
- **5-minute presentation & demo** (+ optional video inside that window) then **5-minute Q&A**.
- Theme prizes: 1st/2nd/3rd per theme (Defense & Security). 1st place also does a 1-minute closing pitch.

5 Ready-to-paste: Project Summary ($\leq$250 words)

Shepherd is a live evacuation platform for Edmonton built for natHacks 2026 Defense & Security. When a wildfire or tornado spreads with the wind, static "leave west" advice fails: roads jam, shelters become unsafe, and people drive into danger. Shepherd connects three surfaces in one system — a commander dashboard, a public phone web app, and a FastAPI backend — so routing and situational awareness stay in sync as conditions change.

We simulate wind-driven wildfire and tornado scenarios on real OpenStreetMap roads. A forecast estimates when fire reaches each place; we close road segments thirty minutes before predicted arrival and score routes as ok, tight, or blocked. The phone shows leave-by times and multi-option routes, then keeps re-routing mid-trip when a path goes unsafe or congested. The dashboard shows hazard growth, closed roads, traffic congestion ribbons, live phone paths, cut-off warnings, and an estimate of how many people and cars need to leave near the hazard pin (zone density $\times$ day/time). Traffic uses a standard BPR delay model with anti-gridlock assignment and projects large crowds (on the order of 100k cars) without freezing the demo.

Judges can join from their own phones via a QR link. Shepherd is a working prototype for emergency decision support and public guidance — not certified life-safety software — with a clear path to agency validation, authenticated roles, and calibrated demand data.

6 Ready-to-paste: About the Project ($\leq$500 words)

Inspiration. Alberta's wildfire seasons and urban evacuation stress made the Defense & Security theme concrete for us: public safety fails when information is static. We wanted one loop where the hazard, the roads, the traffic, the phones, and the commanders all update together — on a real city (Edmonton), not a toy grid.

How we built it. Five people owned clear surfaces under a shared API contract: FastAPI backend (in-memory state),

wildfire/tornado simulation with wind and forecast, OSM-based routing with fire timing and BPR traffic, a MapLibre command dashboard, and a phone web app. We used React/Vite, MapLibre + OSM tiles, OSMnx/NetworkX, SciPy/NumPy/Shapely, and Cloudflare tunnels so guest phones can join the demo. GeoJSON is always [lon, lat]. No database or auth by design for hackathon speed; state lives in memory and resets cleanly.

What we learned. Integration beats isolated features: a beautiful fire model is useless if phones don't reroute and commanders can't see closed roads. Safety margins must be explicit (close at arrival − 30). Large traffic demos need projection (many cars represented by fewer agents) or the API hangs. UX matters under stress — leave-by countdown, collapsed panels, simulated drive by default with a real GPS start pin, and honest copy when no route exists.

Challenges. Syncing five workstreams without breaking `main`; keeping /route fast under crowd assignment; painting continuous traffic corridors on a graph; mid-trip reroutes visible on both phone and dashboard; wind direction conventions (FROM degrees); and staying demo-reliable with tunnels/QR. Flood remains deferred; demand uses transparent zone tables rather than live census — we say that openly.

Next. Calibrated open-census demand, WebSocket push, agency roles and retention limits, live shelter intake, and partnerships with emergency managers to validate the routing rules before any operational use.

7 Suggested 5-minute pitch + demo script

0:00–0:40 Problem + pitch. Alberta/Edmonton. Static advice fails. One-liner + 15-second pitch. Theme: emergency response & public safety.

0:40–1:20 What it is. Three surfaces. Real OSM. Road rule in one sentence. Team of five / contract.

1:20–4:00 Live demo (practice once).

1. Open dashboard Drill · Wildfire · wind west→east · **1×**. Phone already open via QR, Simulated drive, pin at real/GPS or tapped location, Start evacuating.
2. Click map near the phone area — show **people/cars demand** estimate.
3. Play — fire grows; forecast ring; traffic ribbons; closed-road count; blue phone route.
4. When path goes bad: phone toast **Rerouted**; dashboard path redraws; cut-off banner.
5. Optional: Traffic Compare (pile-up vs anti-gridlock). Point at leave-by / ok–tight–blocked.

4:00–4:40 Impact + honesty + next. Prototype not certified. Seed funding → validate with agencies, census demand, auth. Why not Google/Waze: hazard-timed closures + shared live state.

4:40–5:00 Close. Restate one-liner. "Happy to go deeper on routing, traffic, or demand in Q&A."

8 Tools & why (Technical / Feasibility)

Tool	Why
FastAPI + Uvicorn	Fast JSON API, clear CORS, easy for both frontends
React + Vite	Rapid UI iteration; dashboard + phone
MapLibre + OSM tiles	Free maps, no Google billing; GeoJSON overlays
OSMnx + NetworkX	Real Edmonton drive graph
SciPy / NumPy / Shapely	Dijkstra, BPR math, fire geometry in metres
OpenWeather + wind sim	Hazard grows with wind; forecast for "what's next"
Cloudflare Tunnel (`./demo.sh`)	Judges join via QR off laptop LAN
Public Sans + Alberta-ish blues	Looks like government ops tooling, not a toy

9 Curveball Q&A (Pitch & Demonstration)

Production / life-safety certified?
No — hackathon prototype. Clear safety margin + real OSM; needs agency validation, auth, persistence before any operational use.

Where do ∼100k cars come from?
Transparent model: zone density × hazard footprint × day/time occupancy → people → cars; projected onto fewer simulated agents so demos stay real-time.

Why not Google Maps / Waze?
Consumer maps don't take our fire forecast, timed road closures, or shelter safety as first-class inputs, and don't give commanders the same live state.

Why only some roads coloured?
Intentional. Only actionable layers: fire, closed roads, jammed corridors, active phone routes.

Flood / other hazards?
Flood deferred in the contract. Same architecture (hazard polygons + road deadlines + reroute) extends.

How accurate is the fire model?
Wind-driven demo/forecast engine for timing logic, not a full firefighting product. Routing uses `arrival_min` and the −30 buffer.

Privacy?
No auth in demo; temporary pins (stale ∼30s); priority stores type + last 4 only. Production needs consent and retention limits.

What's novel?
End-to-end loop on real Edmonton OSM: wind hazard → timed closures → live phone reroute → commander cut-offs and demand.

Interdisciplinary / team?
Five builders, five surfaces (API, routing, sim, dashboard, phone), one contract — software engineering + emergency UX + traffic modelling.

10 Team & ownership (for credits / bonus)

Harsit	`backend/api`, `shared/`, integration, pitch
Azer	`backend/routing`
Dviviz	`backend/sim`
Shriyaz	`dashboard/`
Aarya	`phone/`

11 If the demo flakes

Refresh both UIs; confirm branch badge; restart `./demo.sh`; use the dashboard QR (not an old join link); Reset before the next run. Fallback narrative: walk closed-road rule + show a recorded clip if you prepared one inside the 5 minutes.

Shepherd · natHacks 2026 · Defense & Security · Internal presenter brief — give this PDF to the slide/pitch owner.